

## SAFETY DATA SHEET

Creation Date 23-March-2012

Revision Date 29-March-2024

Revision Number 3

### 1. Identification

**Product Name** Tetra-n-butylammonium borohydride

**Cat No. :** L12765

**CAS-No** 33725-74-5  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                        |              |                    |
|------------------------------------------------------------------------|--------------|--------------------|
| Substances/mixtures which, in contact with water, emit flammable gases | Category 2   | Gas(es) = Hydrogen |
| Skin Corrosion/Irritation                                              | Category 1 B |                    |
| Serious Eye Damage/Eye Irritation                                      | Category 1   |                    |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

In contact with water releases flammable gas  
Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

Do not allow contact with water

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Wear protective gloves/protective clothing/eye protection/face protection

Handle and store contents under inert gas. Protect from moisture

**Response**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Wash contaminated clothing before reuse

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

**Storage**

Store locked up

Store in a dry place. Store in a closed container

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                             | CAS-No     | Weight % |
|-------------------------------------------------------|------------|----------|
| 1-Butanaminium, N,N,N-tributyl-, tetrahydroborate(1-) | 33725-74-5 | >95      |

### 4. First-aid measures

**General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

**Inhalation**

Remove to fresh air. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If not breathing, give artificial respiration.

**Ingestion**

Immediate medical attention is required. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water.

**Most important symptoms/effects**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should

## Notes to Physician

be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation  
Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available

**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

The product causes burns of eyes, skin and mucous membranes. Contact with water liberates extremely flammable gases.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen. Oxides of boron. Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

**Health**  
3

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
W

## 6. Accidental release measures

**Personal Precautions** Evacuate personnel to safe areas. Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

**Environmental Precautions** Do not allow material to contaminate ground water system. Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Do not breathe dust. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep from any possible contact with water. Store under an inert atmosphere. Incompatible Materials. Acids. Water. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Neoprene       | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

A NIOSH/MSHA approved air purifying dust or mist respirator or European Standard EN 149.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                               |                                 |
|-----------------------------------------------|---------------------------------|
| <b>Physical State</b>                         | Powder Solid                    |
| <b>Appearance</b>                             | White                           |
| <b>Odor</b>                                   | No information available        |
| <b>Odor Threshold</b>                         | No information available        |
| <b>pH</b>                                     | No information available        |
| <b>Melting Point/Range</b>                    | 126 - 131 °C / 258.8 - 267.8 °F |
| <b>Boiling Point/Range</b>                    | No information available        |
| <b>Flash Point</b>                            | No information available        |
| <b>Evaporation Rate</b>                       | Not applicable                  |
| <b>Flammability (solid,gas)</b>               | No information available        |
| <b>Flammability or explosive limits</b>       |                                 |
| <b>Upper</b>                                  | No data available               |
| <b>Lower</b>                                  | No data available               |
| <b>Vapor Pressure</b>                         | No information available        |
| <b>Vapor Density</b>                          | Not applicable                  |
| <b>Specific Gravity</b>                       | No information available        |
| <b>Solubility</b>                             | No information available        |
| <b>Partition coefficient; n-octanol/water</b> | No data available               |
| <b>Autoignition Temperature</b>               | No information available        |
| <b>Decomposition Temperature</b>              | 140 °C                          |
| <b>Viscosity</b>                              | Not applicable                  |
| <b>Molecular Formula</b>                      | C16 H40 B N                     |
| <b>Molecular Weight</b>                       | 257.3                           |

## 10. Stability and reactivity

|                                         |                                                                                                                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                                                                                                                 |
| <b>Stability</b>                        | Stable under normal conditions. Moisture sensitive.                                                                                                                                 |
| <b>Conditions to Avoid</b>              | Excess heat. Incompatible products. Exposure to moist air or water.                                                                                                                 |
| <b>Incompatible Materials</b>           | Acids, Water, Strong oxidizing agents                                                                                                                                               |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen, Oxides of boron, Thermal decomposition can lead to release of irritating gases and vapors |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                            |
| <b>Hazardous Reactions</b>              | Contact with water liberates extremely flammable gases.                                                                                                                             |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

#### **Component Information**

**Toxicologically Synergistic** No information available

#### **Products**

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                   | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| 1-Butanaminium,<br>N,N,N-tributyl-,<br>tetrahydroborate(1-) | 33725-74-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

|                                      |                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Reacts with water Persistence is unlikely based on information available.                                      |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                                                                      |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. Is not likely mobile in the environment. |

### 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 14. Transport information

#### DOT

|                             |                                                       |
|-----------------------------|-------------------------------------------------------|
| <b>UN-No</b>                | UN2813                                                |
| <b>Proper Shipping Name</b> | consumer commodity                                    |
| <b>Technical Name</b>       | 1-Butanaminium, N,N,N-tributyl-, tetrahydroborate(1-) |
| <b>Hazard Class</b>         | 4.3                                                   |
| <b>Packing Group</b>        | II                                                    |

#### TDG

|                      |        |
|----------------------|--------|
| <b>UN-No</b>         | UN2813 |
| <b>Hazard Class</b>  | 4.3    |
| <b>Packing Group</b> | II     |

#### IATA

|                                |                                         |
|--------------------------------|-----------------------------------------|
| <b>UN-No</b>                   | UN3131                                  |
| <b>Proper Shipping Name</b>    | Water-reactive solid, corrosive, n.o.s. |
| <b>Hazard Class</b>            | 4.3                                     |
| <b>Subsidiary Hazard Class</b> | 8                                       |
| <b>Packing Group</b>           | II                                      |

#### IMDG/IMO

|                                |                                         |
|--------------------------------|-----------------------------------------|
| <b>UN-No</b>                   | UN3131                                  |
| <b>Proper Shipping Name</b>    | Water-reactive solid, corrosive, n.o.s. |
| <b>Hazard Class</b>            | 4.3                                     |
| <b>Subsidiary Hazard Class</b> | 8                                       |
| <b>Packing Group</b>           | II                                      |

### 15. Regulatory information

#### International Inventories

| Component                                             | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1-Butanaminium, N,N,N-tributyl-, tetrahydroborate(1-) | 33725-74-5 | -   | X    | X    | ACTIVE                                        | 251-658-2 | -      | -   |

| Component                                             | CAS-No     | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------------------------------------|------------|-------|------|------|------|------|------|-------|-------|
| 1-Butanaminium, N,N,N-tributyl-, tetrahydroborate(1-) | 33725-74-5 | X     | -    | -    | -    | X    | -    | -     | -     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

Authorisation/Restrictions according to EU REACH Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                             | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 1-Butanaminium, N,N,N-tributyl-, tetrahydroborate(1-) | 33725-74-5 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                             | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 1-Butanaminium, N,N,N-tributyl-, tetrahydroborate(1-) | 33725-74-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By** Product Safety Department  
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www.thermofisher.com

**Creation Date** 23-March-2012  
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**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

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**End of SDS**